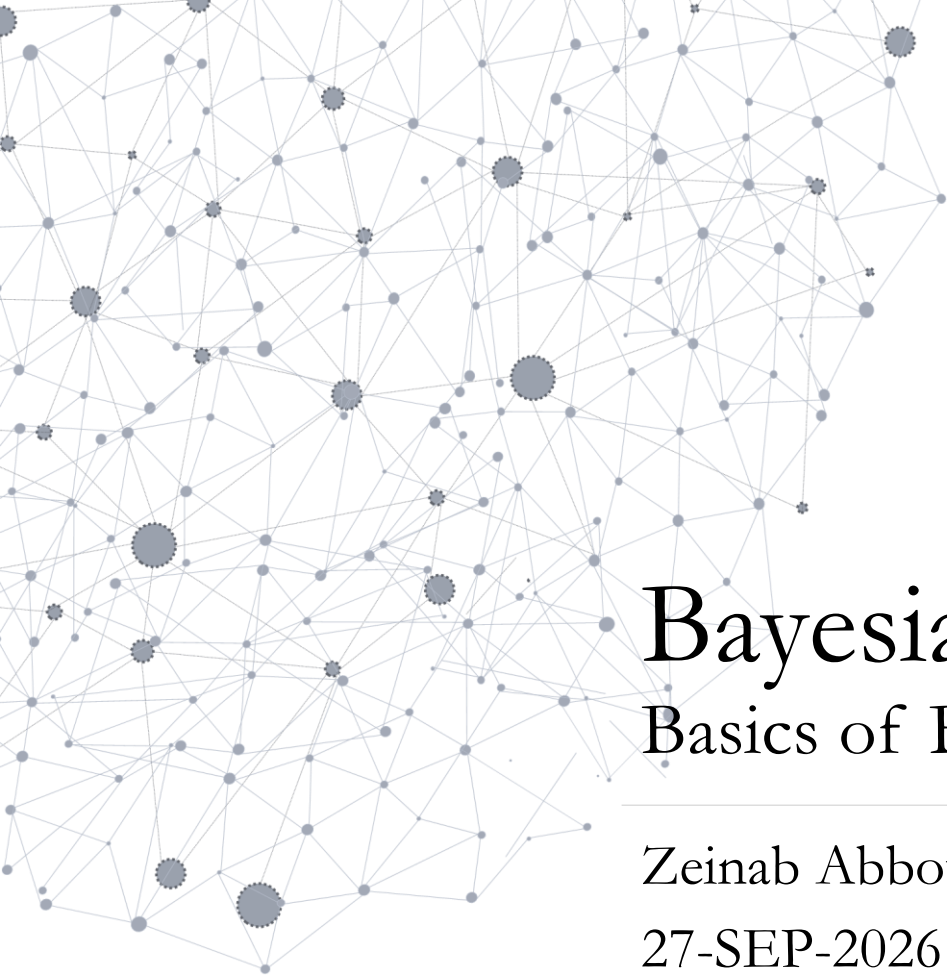




# Bayesian Neural Networks

## Basics of Bayesian Models of Uncertainty

---

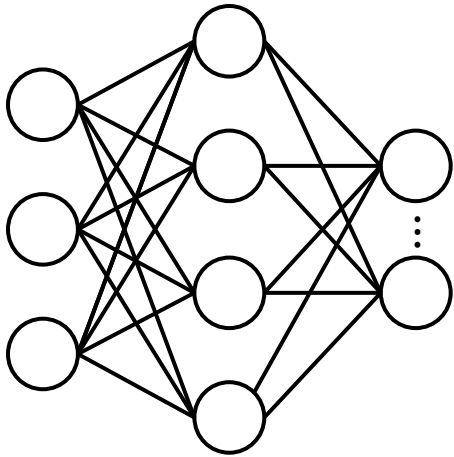
Zeinab Abboud

27-SEP-2026

# Problem with deterministic neural networks

# Problem with deterministic neural networks

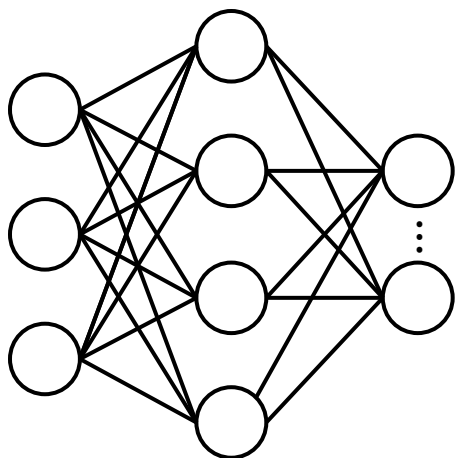
MLP Trained on MNIST



Test Accuracy  $\cong 99\%$

# Problem with deterministic neural networks

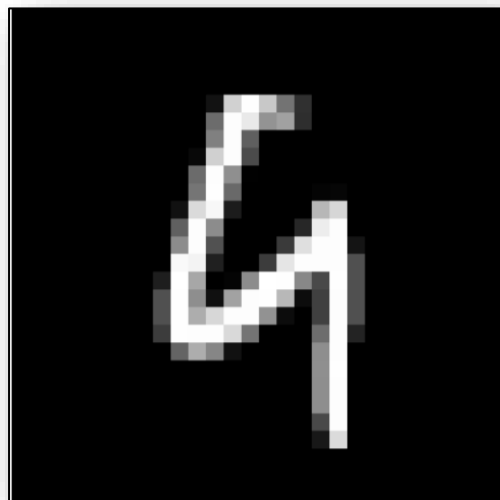
MLP Trained on MNIST



0 1 2 3 4 5 6 7 8 9

Test Accuracy  $\cong 99\%$

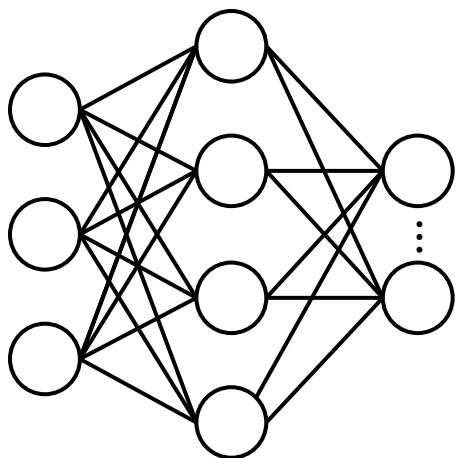
Ambiguous input image



Is it a 4 ? a 9?

# Problem with deterministic neural networks

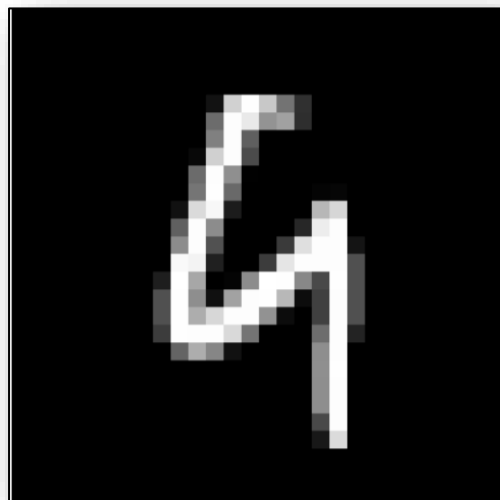
MLP Trained on MNIST



0 1 2 3 4 5 6 7 8 9

Test Accuracy  $\cong 99\%$

Ambiguous input image

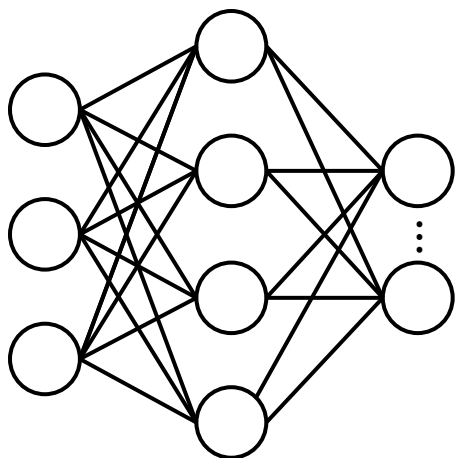


Is it a 4 ? a 9?

MLP model predicts  
 $p(4|\text{image}) = 0.99$

# Problem with deterministic neural networks

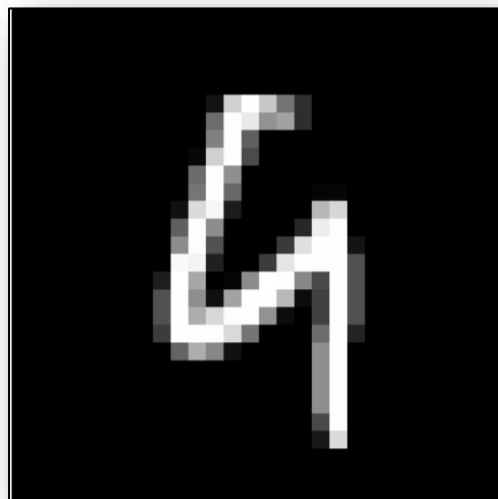
MLP Trained on MNIST



0 1 2 3 4 5 6 7 8 9

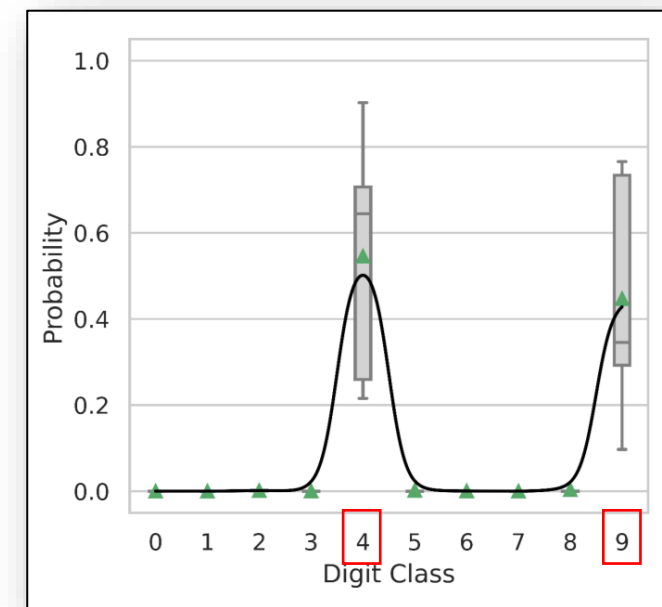
Test Accuracy  $\cong 99\%$

Ambiguous input image



Is it a 4 ? a 9?

MLP model predicts  
 $p(4|\text{4}) = 0.99$



A more realistic  
output of  $p(y|x)$

# Outline

- Recap of deterministic neural network
- Bayes' theorem
- Application of Bayes' theorem in NN
- Hands-on demo

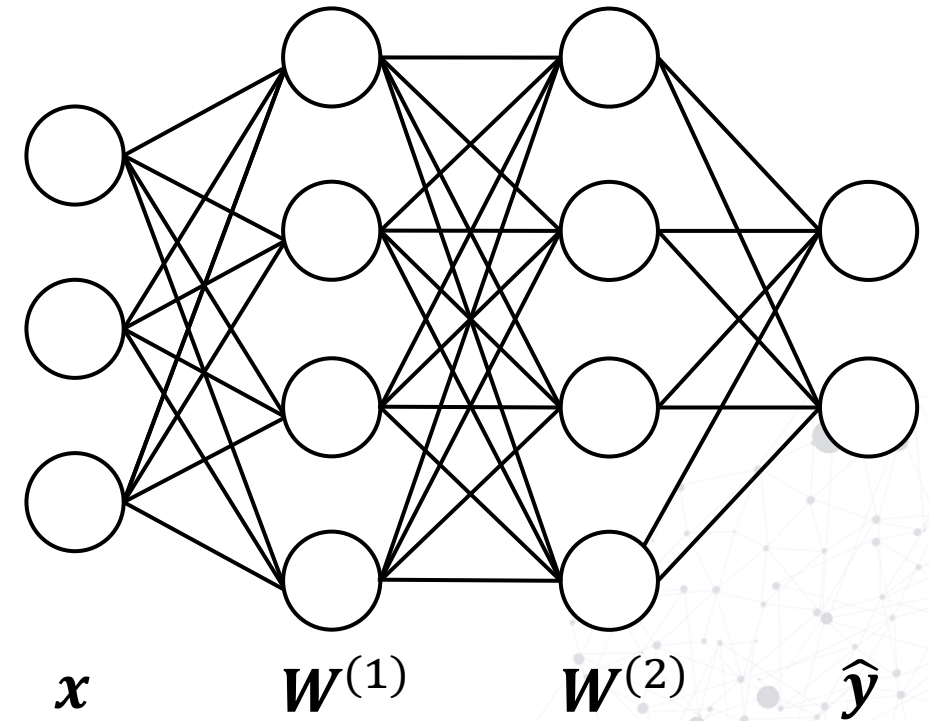
# Neural Networks ....back to basics...

Data  $\rightarrow D = \mathbf{x}^n, \mathbf{y}^n$

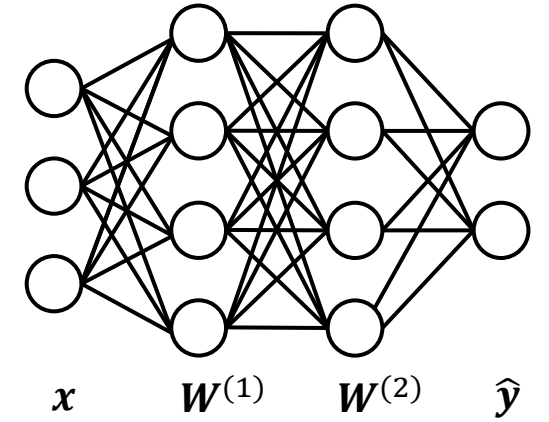
Parameters  $\rightarrow \theta = \{w, b\}$

Model  $\rightarrow p(\mathbf{y}|\mathbf{x}, \theta) \leftarrow$  likelihood

Output  $\rightarrow \hat{\mathbf{y}} = W^{(2)}\sigma(W^{(1)}\mathbf{x} + b)$



# Neural Networks ....back to basics...



Data  $\rightarrow D = x^n, y^n$

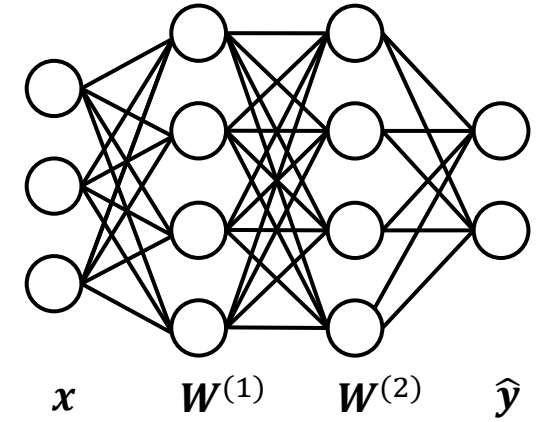
Parameters  $\rightarrow \theta = \{w, b\}$

Model  $\rightarrow p(y|x, \theta)$

Output  $\rightarrow \hat{y} = W^{(2)} \sigma(W^{(1)}x + b)$

$$\hat{y} = \underbrace{\begin{bmatrix} w_{1H}^{(2)} & \dots & w_{1H}^{(2)} \\ w_{H1}^{(2)} & \dots & w_{HN}^{(2)} \end{bmatrix}}_{W^{(2)}} \sigma \left( \underbrace{\begin{bmatrix} w_{11}^{(1)} & \dots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \dots & w_{HN}^{(1)} \end{bmatrix}}_{W^{(1)}} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}}_{x} + \underbrace{\begin{bmatrix} b_{11} \\ \vdots \\ b_{H1} \end{bmatrix}}_{b} \right)$$

# Neural Networks ....back to basics...



Data  $\rightarrow D = x^n, y^n$

Parameters  $\rightarrow \theta = \{w, b\}$

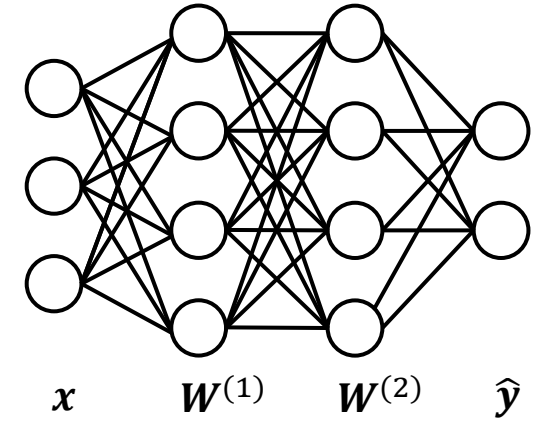
Model  $\rightarrow p(y|x, \theta)$

Output  $\rightarrow \hat{y} = W^{(2)} \sigma(W^{(1)}x + b)$

Point estimates

$$\hat{y} = \underbrace{\begin{bmatrix} w_{1H}^{(2)} & \dots & w_{1H}^{(2)} \\ \vdots & & \vdots \\ w_{H1}^{(2)} & \dots & w_{HN}^{(2)} \end{bmatrix}}_{W^{(2)}} \sigma \left( \underbrace{\begin{bmatrix} w_{11}^{(1)} & \dots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \dots & w_{HN}^{(1)} \end{bmatrix}}_{W^{(1)}} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}}_{x} + \underbrace{\begin{bmatrix} b_{11} \\ \vdots \\ b_{H1} \end{bmatrix}}_{b} \right)$$

# Neural Networks ....back to basics...



Data  $\rightarrow D = x^n, y^n$

Parameters  $\rightarrow \theta = \{w, b\}$

Model  $\rightarrow p(y|x, \theta)$

Output  $\rightarrow \hat{y} = W^{(2)} \sigma(W^{(1)}x + b)$

Point estimates

$$\hat{y} = \underbrace{[0.01 \quad \dots \quad 0.04]_{(1,H)}}_{W^{(2)}} \sigma \left( \underbrace{\begin{bmatrix} 0.01 & \dots & 0.03 \\ \vdots & \ddots & \vdots \\ 0.02 & \dots & 0.02 \end{bmatrix}_{(H,N)}}_{W^{(1)}} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}_{(N,1)}}_x + \underbrace{\begin{bmatrix} 0.1 \\ \vdots \\ 0.2 \end{bmatrix}_{(H,1)}}_b \right)$$

Fixed output  $y$   
given input  $x$

# Let's re-imagine the weights as distributions

$$\begin{aligned}
 & \begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)} \\
 & \quad \uparrow \\
 \hat{y} &= \underbrace{\begin{bmatrix} w_{1H}^{(2)} & \cdots & w_{1H}^{(2)} \end{bmatrix}_{(1,H)}}_{W^{(2)}} \sigma \left( \underbrace{\begin{bmatrix} w_{11}^{(1)} & \cdots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \cdots & w_{HN}^{(1)} \end{bmatrix}_{(H,N)}}_{W^{(1)}} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}_{(N,1)}}_x + \underbrace{\begin{bmatrix} b_{11} \\ \vdots \\ b_{H1} \end{bmatrix}_{(H,1)}}_b \right)
 \end{aligned}$$

# Let's re-imagine the weights as distributions

$$\begin{aligned}
 & \begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)} \\
 & \quad \swarrow \quad \searrow \\
 & \begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix}_{(H,N)} \quad \begin{bmatrix} \sigma_{11}^{2(1)} & \cdots & \sigma_{1N}^{2(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{2(1)} & \cdots & \sigma_{HN}^{2(1)} \end{bmatrix}_{(H,N)} \\
 & \quad \quad \quad \text{Learn the parameters of the distribution} \\
 & \quad \quad \quad \downarrow \\
 & \hat{y} = \underbrace{\begin{bmatrix} w_{1H}^{(2)} & \cdots & w_{1H}^{(2)} \end{bmatrix}_{(1,H)}}_{W^{(2)}} \sigma \left( \underbrace{\begin{bmatrix} w_{11}^{(1)} & \cdots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \cdots & w_{HN}^{(1)} \end{bmatrix}_{(H,N)}}_{W^{(1)}} \underbrace{\begin{bmatrix} x_1^{(1)} \\ \vdots \\ x_N^{(1)} \end{bmatrix}_{(N,1)}}_x + \underbrace{\begin{bmatrix} b_{11} \\ \vdots \\ b_{H1} \end{bmatrix}_{(H,1)}}_b \right)
 \end{aligned}$$

# Let's re-imagine the weights as distributions

$$\begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)}$$

*Learn the parameters of the distribution*

$$\begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix}_{(H,N)}$$

$$\begin{bmatrix} \sigma_{11}^{2(1)} & \cdots & \sigma_{1N}^{2(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{2(1)} & \cdots & \sigma_{HN}^{2(1)} \end{bmatrix}_{(H,N)}$$

Now what?

How do we learn these parameters?

# Bayesian Neural Networks

- Rethink of the neural network as a method to solve Bayes' theorem

$$p(\theta|x, y) = \frac{p(y|x, \theta) p(\theta)}{p(D)}$$

- Learn a posterior distribution  $p(\theta|x, y)$  given a prior  $p(\theta)$  over the neural network parameters.

# Bayesian Neural Networks

- Rethink of the neural network as a method to solve Bayes' theorem

$$p(\theta|x, y) = \frac{p(y|x, \theta) p(\theta)}{p(D)}$$

- Learn a posterior distribution  $p(\theta|x, y)$  given a prior  $p(\theta)$  over the neural network parameters.

What does learning a posterior mean?

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$



# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's break this down...

From Axioms of probability:

$$p(A \cap B) = p(B|A)p(A)$$



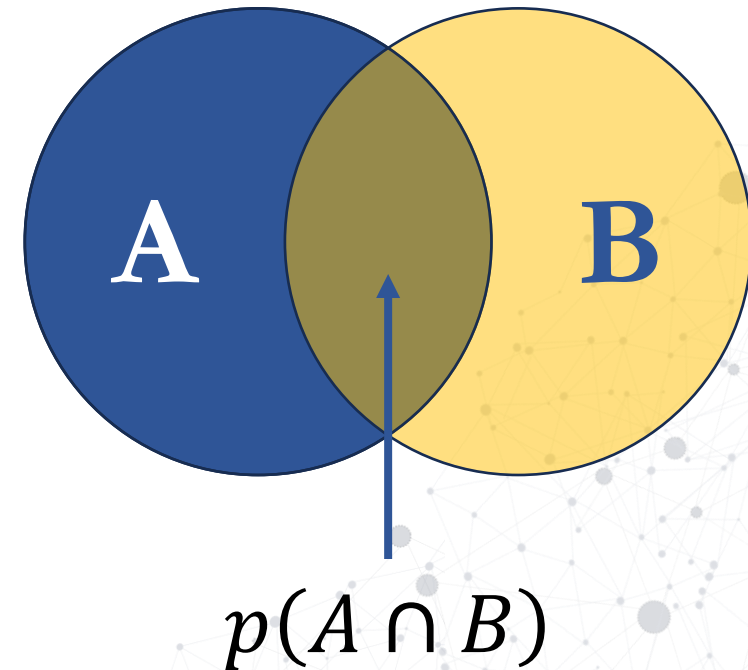
# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's break this down...

From Axioms of probability:

$$p(A \cap B) = p(B|A)p(A)$$



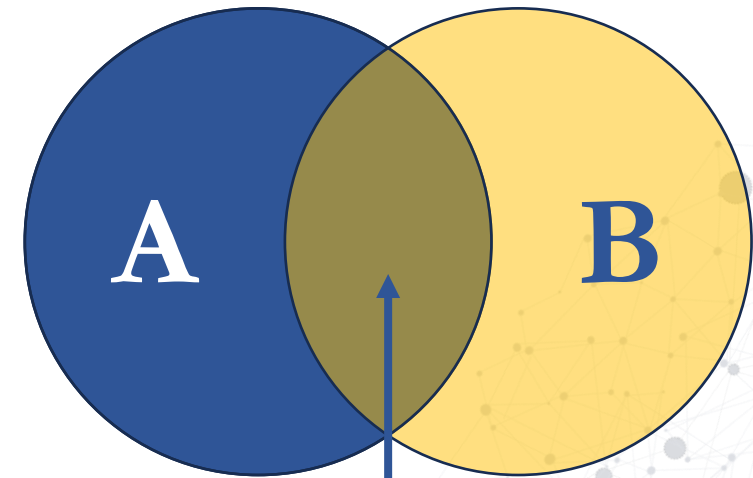
# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's break this down...

From Axioms of probability:

$$p(A \cap B) = p(B|A)p(A)$$



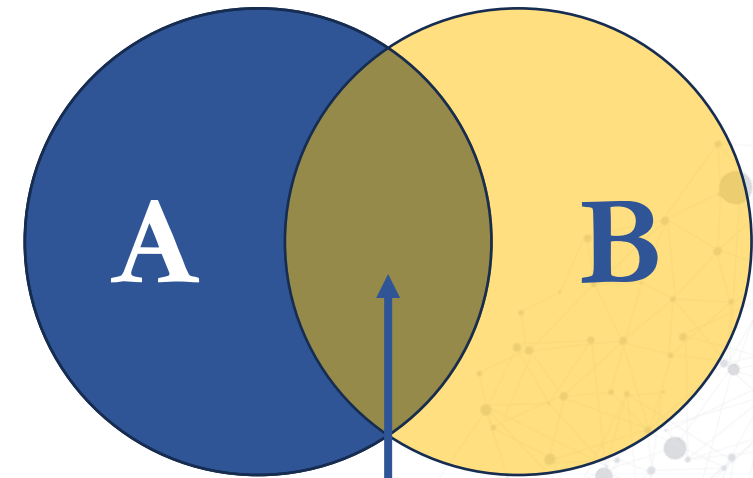
$$p(A \cap B) \text{ or } p(B \cap A) \\ \text{or } p(A, B)$$

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's break this down...

$$p(A \cap B) = p(B|A)p(A)$$
$$p(B \cap A) = p(A|B)p(B)$$



$p(A \cap B)$  or  $p(B \cap A)$   
or  $p(A, B)$

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

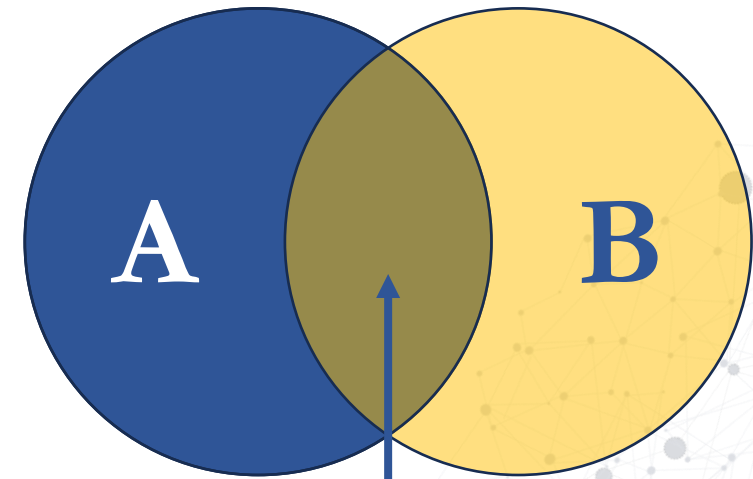
Let's break this down...

$$p(A \cap B) = p(B|A)p(A) \cdots (1)$$

$$p(B \cap A) = p(A|B)p(B) \cdots (2)$$

$$\therefore (1) = (2)$$

$$\therefore p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$



$p(A \cap B)$  or  $p(B \cap A)$   
or  $p(A, B)$

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease 

Precision of the test is 99%

Prevalence of  in population is 0.1%

Q: what is the probability that you have disease   
given that you tested positive?

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease 

Precision of the test is 99%

Prevalence of  in population is 0.1%

Q: what is the probability that you have disease   
given that you tested positive?  $p(A|B)$

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease 

Precision of the test is 99%

Prevalence of  in population is 0.1%  $p(A)$

Q: what is the probability that you have disease   
given that you tested positive?  $p(A|B)$

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease 

Precision of the test is 99%  $p(B|A)$

Prevalence of  in population is 0.1%  $p(A)$

Q: what is the probability that you have disease   
given that you tested positive?  $p(A|B)$

# Bayes' Theorem

$$p(B) = p(B|A)p(A) + p(B|\sim A)p(\sim A)$$

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease   $p(B)$

Precision of the test is 99%  $p(B|A)$

Prevalence of  in population is 0.1%  $p(A)$

Q: what is the probability that you have disease   
given that you tested positive?  $p(A|B)$

# Bayes' Theorem

$$p(B) = p(B|A)p(A) + p(B|\sim A)p(\sim A)$$

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)} = \boxed{0.09}$$

Let's say you took a medical test, and tested positive for disease   $p(B)$

Precision of the test is 99%  $p(B|A)$

Prevalence of  in population is 0.1%  $p(A)$

Q: what is the probability that you have disease   
given that you tested positive?  $p(A|B)$

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

Let's say you took a medical test, and tested positive for disease  $p(B)$

Precision of the test is 99%  $p(B|A)$

Prevalence of the disease in population is 0.1%  $p(A)$

Q: what is the probability that you have disease  
given that you tested positive?  $p(A|B)$

**Now let's say you took another test to confirm whether you *actually* do have the disease or whether the first one was a false positive**

# Bayes' Theorem

$$p_2(A|B) = \frac{p(B|A)p_1(A|B)}{p_1(B)} = 0.91$$

Let's say you took a medical test, and tested positive for disease  $p(B)$

Precision of the test is 99%  $p(B|A)$

Prevalence of the disease in population is 0.1%  $p(A) = p_1(A|B) = 0.09$

Q: what is the probability that you have disease  
given that you tested positive?  $p(A|B)$

**Now let's say you took another test to confirm whether you *actually* do have the disease or whether the first one was a false positive**

Now that we have more evidence, we have a more informed prior, which is the posterior we previously calculated..

# Bayes' Theorem

$$p_3(A|B) = \frac{p(B|A)p_2(A|B)}{p_2(B)} = 0.999$$

Let's say you took a medical test, and tested positive for disease  $p(B)$

Precision of the test is 99%  $p(B|A)$

Prevalence of the disease in population is 0.1%  $p(A) = p_2(A|B) = 0.01$

Q: what is the probability that you have disease  
given that you tested positive?  $p(A|B)$

**Now let's say you took another test to confirm whether you *actually* do have the disease or whether the first one was a false positive**

Now that we have more evidence, we have a more informed prior, which is the posterior we previously calculated..

**Let's say you're still doubtful that you have the disease, you take another test, and it's positive...**

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A)p_{i-1}(A|B)}{p(B)} = 0.91$$

Let's say you took a medical test, and tested positive for disease  $p(B)$

Bayes' theorem allows us to update the conditional probability of an event occurring when new data is presented

Now let's say you took another test to confirm whether you *actually* do have the disease or whether the first one was a false positive

Now that we have more evidence, we have a more informed prior, which is the posterior we previously calculated..

# Bayes' Theorem

$$p(A|B) = \frac{p(B|A) p(A)}{p(B)}$$

Posterior

Likelihood Prior

Evidence  
(Marginal)

# Bayes' Rule for Neural Networks

$$p(\theta|x, y) = \frac{\overset{\text{Likelihood}}{p(y|x, \theta)} \overset{\text{Prior}}{p(\theta)}}{\underset{\substack{\text{Evidence} \\ \text{(Marginal)}}}{p(D)}}$$

Posterior

# Bayes' Theorem

$$p(A|B) = \frac{\overset{\text{Likelihood}}{p(B|A)} \overset{\text{Prior}}{p(A)}}{\underset{\substack{\text{Evidence} \\ \text{(Marginal)}}}{p(B)}}$$

Posterior

# Bayes' Rule for Neural Networks

$$p(\theta|x, y) = \frac{\overset{\text{Likelihood}}{p(y|x, \theta)} \overset{\text{Prior}}{p(\theta)}}{\underset{\text{Evidence (Marginal)}}{p(D)}}$$

Posterior

Evidence  
(Marginal)

$$p(D) = \int p(D|\theta)p(\theta)d\theta$$

In high dimensional parameter space  $\theta$  this integral is intractable (i.e. incomputable)

$$p(D) = \int_{\theta_0} \cdots \int_{\theta_{D-1}} p(X, \theta) d\theta_0 \cdots d\theta_{D-1}$$

# Variational Inference for BNNs

*approximating a posterior*

$$p(\theta|x, y) = \frac{\overset{\text{Likelihood}}{p(y|x, \theta)} \overset{\text{Prior}}{p(\theta)}}{\underset{\text{Evidence (Marginal)}}{p(D)}}$$

Posterior

Evidence  
(Marginal)



$$q_{\phi}(\theta) \approx p(\theta|x, y)$$

Approximate  
Posterior

# Variational Inference for BNNs

*approximating a posterior*

- Introduce a tractable family of distributions  $q_{\phi}(\theta) = \prod q(\theta_i)$  (\*)  
Diagonal Gaussian with mean  $\mu$  and variance  $\sigma^2$
- Now the optimization criterion becomes of the following form

$$\mathcal{L}_{ELBO} = -\mathbb{E}[\log p(y|x, \theta)] + \beta \cdot D_{KL}(q_{\phi}(\theta) || p(\theta))$$

Likelihood  
Data-dependent

KL divergence  
Prior-dependent

\*with mean-field assumption, which assumes that all the parameters within the network are independent random variables

# Let's re-imagine the weights as distributions

$$\begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)}$$

$\rightarrow$ 

$$\begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix}_{(H,N)}$$
*Learn the parameters of the distribution*

$\rightarrow$ 

$$\begin{bmatrix} \sigma_{11}^{2(1)} & \cdots & \sigma_{1N}^{2(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{2(1)} & \cdots & \sigma_{HN}^{2(1)} \end{bmatrix}_{(H,N)}$$

Now what?

How do we learn these parameters?

$\hat{y}$

$W^{(2)}$

$W^{(1)}$

$x$

$b$

$(H,1)$

# How do we learn these distributions?

$$\begin{bmatrix} \mathcal{N}_{11}(\mu_{11}, \sigma_{11}^2) & \cdots & \mathcal{N}_{1N}(\mu_{1N}, \sigma_{1N}^2) \\ \vdots & \ddots & \vdots \\ \mathcal{N}_{H1}(\mu_{H1}, \sigma_{H1}^2) & \cdots & \mathcal{N}_{HN}(\mu_{HN}, \sigma_{HN}^2) \end{bmatrix}_{(H,N)} \begin{matrix} \rightarrow \\ \rightarrow \end{matrix} \begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix}_{(H,N)} \begin{matrix} \text{Learn the} \\ \text{parameters of} \\ \text{the distribution} \end{matrix}$$

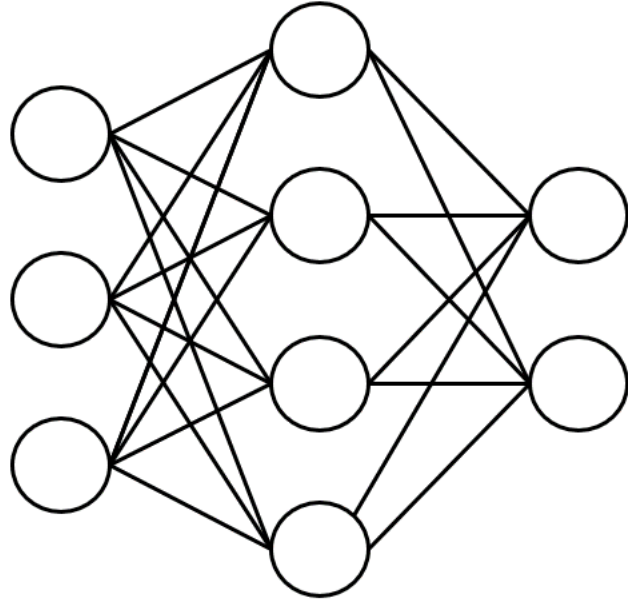
$$\begin{bmatrix} \sigma_{11}^{2(1)} & \cdots & \sigma_{1N}^{2(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{2(1)} & \cdots & \sigma_{HN}^{2(1)} \end{bmatrix}_{(H,N)}$$

Reparameterization Trick:  $w_{ij} = \mu_{ij} + \sigma_{ij}\epsilon ; \epsilon \sim \mathcal{N}(0, 1)$

$$\begin{bmatrix} w_{11}^{(1)} & \cdots & w_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{H1}^{(1)} & \cdots & w_{HN}^{(1)} \end{bmatrix}_{(H,N)} = \begin{bmatrix} \mu_{11}^{(1)} & \cdots & \mu_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \mu_{H1}^{(1)} & \cdots & \mu_{HN}^{(1)} \end{bmatrix} + \begin{bmatrix} \sigma_{11}^{(1)} & \cdots & \sigma_{1N}^{(1)} \\ \vdots & \ddots & \vdots \\ \sigma_{H1}^{(1)} & \cdots & \sigma_{HN}^{(1)} \end{bmatrix} \cdot \epsilon$$

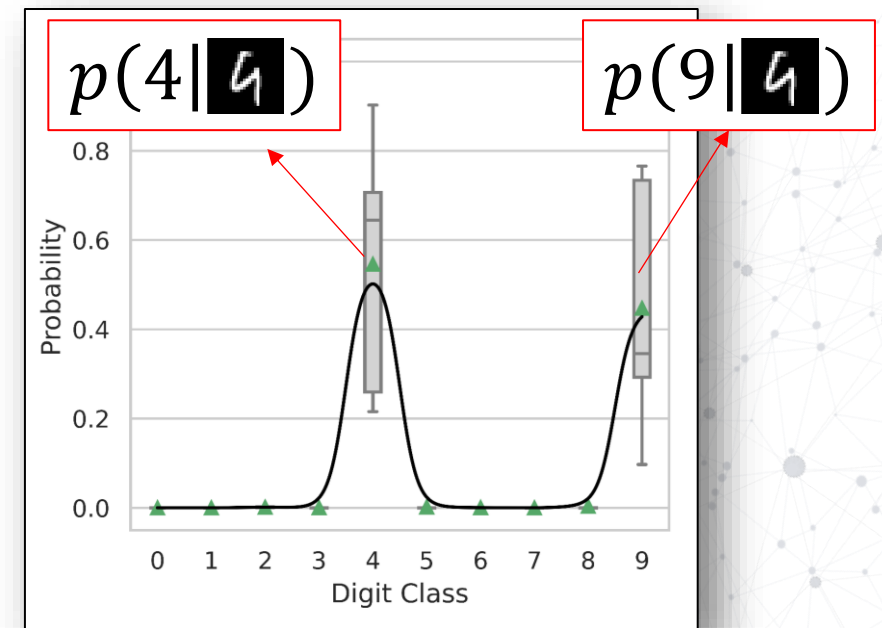
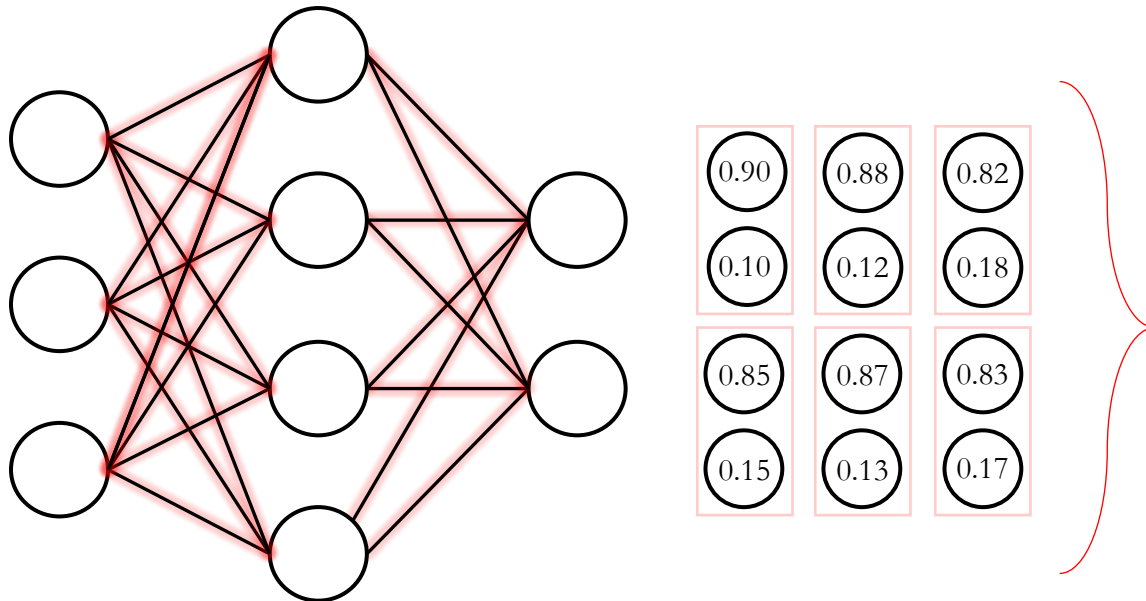
# Inference & Estimating Uncertainty

- At inference time, you run the feedforward network for each sample  $N$  times, effectively thought of a  $N$  ensembles of neural networks
- At each feedforward step:  $w_{ij}^l = \mu_{ij}^l + \sigma_{ij}^l \epsilon ; \epsilon \sim \mathcal{N}(0, 1)$



# Inference & Estimating Uncertainty

- At inference time, you run the feedforward network for each sample  $N$  times, effectively thought of a  $N$  ensembles of neural networks
- At each feedforward step:  $w_{ij}^l = \mu_{ij}^l + \sigma_{ij}^l \epsilon$  ;  $\epsilon \sim \mathcal{N}(0, 1)$



# Other benefits...

- Variational continual learning (ICLR 2018) for sequential multiclass, multitask learning
  - Updating the prior based on the posterior of the previously learned task/class

$$\mathcal{L}_{ELBO} = -\mathbb{E}[\log p(y|x, \theta)] + \beta \cdot D_{KL}(q_{\phi}(\theta) || p(\theta))$$

$$\mathcal{L}_{VCL}^t(q_t(\theta)) = -\mathbb{E}[\log p(y|x, \theta)] + \beta \cdot D_{KL}(q_t(\theta) || q_{t-1}(\theta))$$

# Other approaches...

Exact Ideal ←—————▶ Scalable Practical

Exact  
MCMC  
(HMC)

SG-MCMC  
(SGLD/SGHMC)

Variational  
Inference

Laplace/  
SWAG

MC Dropout/  
Deep Ensembles

Additional References:

- [Pattern Recognition & ML](#), 3.3, 4.4, 5.7 – Bishop 2006
- [Machine Learning: A probabilistic perspective](#) – Kevin Murphy 2012
- [Probabilistic Artificial Intelligence](#) – Krause & Hubotter 2025
- Introduction to Bayesian Inference & Decision – Robert L. Winkler 1972
- [Bayesian Learning for Neural Networks](#) – R. M. Neal 1995